EE 242 Lab 0 – Texts, Code, Plot and Play

In this lab exercise, basic python tools that will be required throughout this course to perform the various lab assignments.

Form Teams

All in-lab sessions will be conducted as a team of 2-4 students. We prefer teams of 3 over other combinations.

Write down the names and number of your team (assigned by the TA of the form "XXYY" where XX is your lab session code (AA, AB, AD, AE, AF) and YY is your team number.

Team XXYY: Name 1, Name 2, Name 3 **DONT FORGET TO PUT YOUR TEAM INFO HERE**

Playing with strings

Lab 0.1 - Palindromes

Write the python code for reading input from user and write a function to check whether the string is a palindrome using the following algorithm

- read input in a string a
- reverse the string b
- write a for loop to check from 1 → N if a[i]=b[i]
- if all characters match return True else return False

```
#Python code goes here
a=input("Input A:")
def isPalindrome(input_string):
    #your code goes here

    letters = list(a)
    reversed_letters = letters[::-1]
    #reversed_word = "".join(reversed_letters)

    for i in range(len(a)):

      if letters[i] != reversed_letters[i]:
        return False;



    return True;

b=isPalindrome("Mahmood Hameed");
print("Output B:"+str(b)) #Should be False
```

```
c=isPalindrome("Signals, Systems and Data I:I ataD dna
smetsyS ,slangiS")
print("Output C:"+str(c)) #Should be True
d=isPalindrome(a)
print("Output D:"+str(d))
```

Questions to be answered in the report:

1. Is there a way to make this algorithm more efficient in time required and storage space? Traversing the string directly instead of creating and storing an entirely new string would save space

2. Are there cases where the above algorithm/code will fail? Main one is an empty string, the length of the input string is 0, which by range(0), means the loop is never used and it automatically returns true regardless of what the palindrom fuction is comparing to

Lab 0.2 - Teams

Write the python code for reading input from user and write a function to replace all vowels using the following encoding scheme:

- Aa → @
- Ee → 3
- Ii → 1
- Oo → 0
- Uu → ^

Show the TA example with all of your names concatenated in the string.

```
#python code goes here
a=input("Input A:")
def encode(input_string):
    #your code goes here

    output_string = ""

    for char in input_string:
        if char == 'a' or char == 'A':
            output_string += '@'
        elif char == 'e' or char == 'E':
            output_string += '3'
        elif char == 'i' or char == 'I':
            output_string += '1'
        elif char == 'o' or char == 'O':
            output_string += '0'
        elif char == 'u' or char == 'U':
            output_string += '^'
        else:
```

```
            output_string += char

    return output_string


b=encode("Mahmood Hameed");
print("Output B:"+b) #Should be M@hm00d H@m33d
c=encode("Signals, Systems and Data I: This is my first lab session")
print("Output C:"+c) #Should be S1gn@ls, Syst3ms @nd D@t@ 1: Th1s 1s
my f1rst l@b s3ss10n
d=encode(a)
print("Output D:"+d)
```

Questions to be answered in the report:

1. What is the time complexity of the code if the length of the string is N? given that there is a fixed number of statements / comparisons that can be made and given that the entire program is capped off at whatever the length of the string is, the time complexity would be O(N) **2. How much storage does your function require if the length of the string is N? (1 char is 1 byte)** Given the existing storage of length N for the string as well as the equally length output string also of length N, the total storage requred would be 2N

Playing with Signals

Lab 0.3 - Reading, playing and writing audio files

Use scipy package to read the provided baby.wav file. **Refer to documentation (https://docs.scipy.org/doc/scipy/reference/io.html)**

Perform these 4 actions :

- Load the wave file
- Plot the wave file in time-domain **Refer to this demo for matplotlib (https://matplotlib.org/stable/plot_types/basic/plot.html#sphx-glr-plot-types-basic-plot-py)**
- Write a new wave file called "babyhigh.wav" with a sample rate of 96000
- Write a new wave file called "babylow.wav" with a sample rate of 24000

Use IPython package to play the provided baby.wav file as well as the other two "babyhigh.wav" and "babylow.wav" files. **Refer to documentation (https://ipython.readthedocs.io/en/stable/api/generated/IPython.display.html)**

```
#Python Code goes here
import matplotlib.pyplot as plt
import numpy as np
from scipy.io import wavfile as wav
import sounddevice as sd
```

```
#Install the IPython package using !python install --user IPython
#1. Read the baby.wav file.
w = wav.read("baby.wav")

#2. Plot the laugh sound (What are the two arrays that you see?)
plt.style.use('_mpl-gallery')

# make data
print(w)

samplerate, data = w
print(f"number of channels = {data.shape[1]}")
length = data.shape[0] / samplerate
print(f"length = {length}s")

# plot
time = np.linspace(0., length, data.shape[0])
plt.plot(time, data[:, 0], label="Left channel")
plt.plot(time, data[:, 1], label="Right channel")
plt.legend()
plt.xlabel("Time [s]")
plt.ylabel("Amplitude")
plt.show()

# make data


#3. Store the data with sample rate of 96000

wav.write("babyhigh.wav", 96000, data)


#4. Store the data with sample rate of 24000

wav.write("babylow.wav", 24000, data)

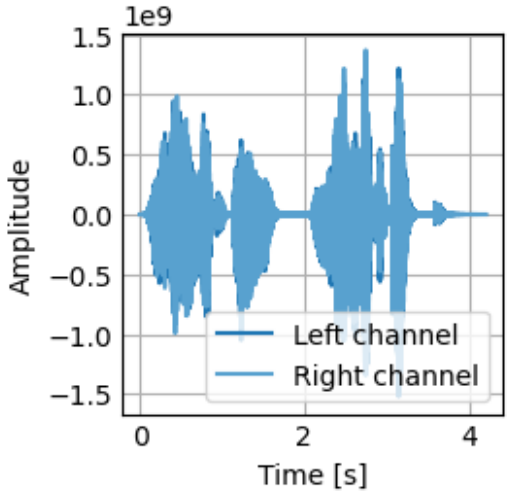

#5. Play all the files. Notice how the sound changes?

fs, data = wav.read("babyhigh.wav")
sd.play(data, fs)
sd.wait()

fs, data = wav.read("babylow.wav")
sd.play(data, fs)
sd.wait()


C:\Users\argames\AppData\Local\Temp\ipykernel_4324\3928992771.py:10:
WavFileWarning: Chunk (non-data) not understood, skipping it.
  w = wav.read("baby.wav")
```

```
(48000, array([[-2816,  2816],
       [ 4096, -2816],
       [-2816,  2816],
       ...,
       [ 1280,     0],
       [    0,  1280],
       [-1280, -1280]], shape=(201600, 2), dtype=int32))
number of channels = 2
length = 4.2s
```



Questions to be answered in this report:

1. Store the wavefile with a lower sampling rate and try to play it. Does it sound higher pitch or lower pitch? (Keep the IPython display open to show TA) It generally plays at a lower pitch

2. Store the wavefile with a higher sampling rate and try to play it. Does it sound higher pitch or lower pitch? (Keep the IPython display open to show TA)

It generally plays at a higher pitch